STANDARD A4 COMPLETE SYSTEM DOSSIER · 103 TABLES EDITION

FARMSKING™ ENTERPRISE ARCHITECTURE

Complete Feature-by-Feature & Table-by-Table Master Specification

This document serves as the authoritative architectural blueprint and developer onboarding bible for FARMSKING TECHNOLOGIES PRIVATE LIMITED. It exhaustively details all 103 relational database models, business logic modules, double-entry financial settlements, multi-vendor marketplace engines, clinical AI agronomist diagnostic bridges, satellite remote-sensing pipelines, and Indian MCA/GST statutory compliance frameworks.

CORPORATE ENTITY **FARMSKING TECHNOLOGIES PRIVATE LIMITED**	CORPORATE IDENTITY NO. (CIN) **U01100PB2026PTC063294**
REGISTRAR OF COMPANIES **Ministry of Corporate Affairs, ROC Chandigarh**	REGISTERED OFFICE **VPO Goindwal Sahib, Tehsil Khadur Sahib, Tarn Taran, Punjab - 143422**
TOTAL VERIFIED TABLES **103 Database Models (100% Timestamped with createdAt/updatedAt)**	TECHNOLOGY ARCHITECTURE **NestJS (TS) · Prisma ORM · PostgreSQL (Neon) · Expo React Native Web**

5. Clinical Agronomy & Doctor Consultation

This architectural pillar manages high-concurrency state transitions, relational data consistency, and operational workflows for **6** core database tables within the FarmsKing ecosystem.

AdvisorAssignment

`Table: advisor_assignments` 16 Fields · advisor_assignment_schema_only_.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Formal professional assignment linking a farmer/gardener to their certified Agronomist/Crop Doctor.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Tracks advisorId, farmerId, subscriptionId, status (PENDING, ACTIVE, REVOKED, ADMIN_APPROVAL_PENDING), and notes.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Governs who has permission to view plot clinical data, write prescriptions, and receive consultation payouts.

CallRequest

`Table: call_requests` 9 Fields · advisor_assignment_schema_only_.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Coordinates scheduled 1-on-1 audio/video consultation appointments between farmer and crop doctor.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores requested time slot, advisorId, farmerId, call status, call duration, and call recording link.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Enables structured remote agronomy advisory without doctors being inundated with random calls.

PartnerAssignment

`Table: partner_assignments` 10 Fields · advisor_assignment_schema_only_.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Territory assignment linking village business partners / franchises to local farmers.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Maps businessPartnerId to farmerId, tracking offline support, VIP pass delivery, and commission attribution.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Maintains strong rural grassroots presence and motivates field partners with commission tracking.

VoiceAILog

`Table: voice_ai_logs` 8 Fields · section_381.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Audit log of conversational Voice AI assistant interactions in regional dialects.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores caller mobile, transcribed audio prompt, AI intent detected, spoken audio reply, and duration.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Enables non-literate farmers to ask questions via simple voice queries in Punjabi/Hindi/English.

GroupVoiceCall

`Table: group_voice_calls` 11 Fields · section_884.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Enables village-wide group audio townhalls and masterclasses conducted by senior agronomists.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Manages live call room, channel token, host agronomist, start/end time, and total listeners.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Allows one expert to broadcast urgent pest management advice to hundreds of farmers simultaneously.

GroupVoiceCallParticipant

`Table: group_voice_call_participants` 10 Fields · section_884.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Participant roster for village audio townhalls.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Tracks participant userId, join time, leave time, hand-raise status, and mute state.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Monitors farmer community engagement and rewards active participants.

3. Precision Agronomy & Farm Parcel Registry

This architectural pillar manages high-concurrency state transitions, relational data consistency, and operational workflows for **33** core database tables within the FarmsKing ecosystem.

ExpenseCategory

`Table: expense_categories` 11 Fields · expenses_sales_customers_payments.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Classification tree for farm expenses (Seeds, Diesel, Fertilizer, Machinery Repair, Lease Rent).

2. How it works ("ਕਿਵੇਂ ਹੈ"): Standardized categories with custom subcategories per farm.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Structures farm financial reports and profit-and-loss statements.

Expense

`Table: expenses` 32 Fields · expenses_sales_customers_payments.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Individual operational expenditure voucher logged on the farm.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores date, amount, categoryId, vendor/party, invoice receipt photo, and related plot/crop.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Calculates total cost of cultivation, helping farmers realize true net profit after harvest.

Party

`Table: parties` 10 Fields · expenses_sales_customers_payments.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Counterparty directory (Local Arhtiya, Mandi Trader, Diesel Pump, Fertilizer Dealer).

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores party name, shop address, mobile, opening balance, and ledger type (RECEIVABLE/PAYABLE).

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Consolidates traditional rural credit and informal credit lines into clear digital accounts.

PartyLedgerEntry

`Table: party_ledger_entries` 13 Fields · expenses_sales_customers_payments.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Running khata/ledger for transactions with external parties.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Records debit, credit, balance, reference voucher, payment mode, and notes.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Eliminates disputes between farmers and commission agents (Arhtiyas) over interest and balances.

PaymentReceipt

`Table: payment_receipts` 12 Fields · expenses_sales_customers_payments.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Digital formal receipt issued for any financial settlement on FarmsKing.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores receiptNumber, grossAmount, deductions, netAmount, payerId, payeeId, and statutory seal.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Provides tamper-proof digital receipts sent via SMS and WhatsApp.

SaleBill

`Table: sale_bills` 23 Fields · expenses_sales_customers_payments.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Produce sale invoice issued by farmer to grain buyers.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores crop sold, quintals, rate per quintal, total amount, buyer name, and payment receipt status.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Generates digital proof of agricultural income required for bank loans and statutory tax exemption.

Sale

`Table: sales` 20 Fields · expenses_sales_customers_payments.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Farm sales transaction header.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Records sale date, gross amount, net amount, customerId, and fulfillment state.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Feeds top-line agricultural revenue into the farmer's profit analytics engine.

SaleItem

`Table: sale_items` 10 Fields · expenses_sales_customers_payments.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Individual produce item in a farm sales invoice.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores cropId, variety, weight, unit rate, and total line price.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Enables commodity-wise revenue comparison across different seasonal crops.

Payment

`Table: payments` 14 Fields · expenses_sales_customers_payments.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Receipt of incoming money from crop buyers.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Records payment date, saleId, amount received, payment mode (NEFT/UPI/Cash), and bank reference.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Closes open sales accounts and reconciles farmer bank deposits.

CropActivitySchedule

`Table: crop_activity_schedules` 17 Fields · farmer_advisor_plan_subscription_schedule_uploads_.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Chronological schedule of farm activities (Sowing, Weeding, Irrigation, First Spray, Harvest).

2. How it works ("ਕਿਵੇਂ ਹੈ"): Generated from agronomy rules or assigned by crop doctor; tracks due date, completion status, and chemical notes.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Guides farmers day-by-day to maximize yield and prevent missed critical agronomic windows.

LabourWorker

`Table: labour_workers` 23 Fields · labour_fertilizer_spray_harvest_inventory_machinery_irrigation.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Master directory of farm labourers, contractual workers, and machine operators.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores worker name, mobile, Aadhaar/ID, daily wage rate, and balance payable.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Maintains ethical labour records and simplifies weekly cash wage settlements.

LabourEntry

`Table: labour_entries` 21 Fields · labour_fertilizer_spray_harvest_inventory_machinery_irrigation.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Daily attendance and wage ledger for farm labour.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Records date, workerId, plotId, operation (e.g. Cotton Picking, Paddy Transplanting), hours, and calculated wage.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Gives farmers exact operational labour cost breakdown per acre.

LabourWorkEntry

`Table: labour_work_entries` 23 Fields · labour_fertilizer_spray_harvest_inventory_machinery_irrigation.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Piece-rate labour tracker (e.g. wage paid per quintal picked or per acre weeded).

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores units completed, contract rate, bonus/deduction, and total payout.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Supports performance-based incentive settlements during peak harvesting seasons.

LabourPayment

`Table: labour_payments` 14 Fields · labour_fertilizer_spray_harvest_inventory_machinery_irrigation.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Financial payment records disbursed to labourers.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Records payment date, workerId, amount paid, payment mode (Cash/Bank), and advance wage adjustment.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Maintains clean worker passbooks and prevents wage dispute friction in rural communities.

Fertilizer

`Table: fertilizers` 12 Fields · labour_fertilizer_spray_harvest_inventory_machinery_irrigation.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Registry of soil nutrients and fertilizers available on the farm (Urea, DAP, MOP, Zinc, Bio-NPK).

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores nutrient percentage (N-P-K-S-Zn), bag weight, brand, and available inventory.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Helps farmers maintain fertilizer balance without excessive nitrogen application.

FertilizerUsage `Table: fertilizer_usage` 17 Fields · labour_fertilizer_spray_harvest_inventory_machinery_irrigation.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Tracks exact nutrient quantities applied to specific plots.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Records application date, plotId, fertilizerId, quantity applied (kg/bags), and basal vs top-dressing mode.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Calculates cost per acre and monitors soil nutrient depletion over successive seasons.

Spray `Table: sprays` 20 Fields · labour_fertilizer_spray_harvest_inventory_machinery_irrigation.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Log of chemical or bio-spray applications executed in the field.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Records spray date, target pest, product names, water volume in litres, tank count, and operator name.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Prevents chemical overdosage, tracks safety waiting periods (PHI) before harvest.

SpraySchedule `Table: spray_schedules` 18 Fields · labour_fertilizer_spray_harvest_inventory_machinery_irrigation.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Planned future spray event recommended by agronomists.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores scheduled date, target stage (e.g., Tillering stage), recommended fungicide/insecticide, and reminder flags.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Sends timely SMS and push reminders to farmers before pest infestation surges.

SprayProductCatalog `Table: spray_product_catalog` 5 Fields · labour_fertilizer_spray_harvest_inventory_machinery_irrigation.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Technical reference database of approved pesticides, insecticides, and fungicides.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores trade name, technical chemical name, CIB&RC registration number, dosage, and antidote information.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Ensures safety compliance and links spray recommendations directly to the store for one-click purchasing.

SprayItemTemplate `Table: spray_item_templates` 11 Fields · labour_fertilizer_spray_harvest_inventory_machinery_irrigation.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Standardized ICAR-aligned spray formulas for different crops.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores stage name, dosage per acre, chemical combinations, and mixing instructions.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Eliminates confusion and standardizes scientific agronomic advisory across the platform.

CropProblem `Table: crop_problems` 23 Fields · labour_fertilizer_spray_harvest_inventory_machinery_irrigation.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Ticket/Case logged when a crop shows disease, pest attack, or deficiency symptoms.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores description, affected area, severity rating, AI preliminary diagnosis, assignedAdvisorId, and resolution status.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Connects farmers directly to specialists for rapid intervention, saving crops from widespread ruin.

CropProblemPhoto `Table: crop_problem_photos` 6 Fields · labour_fertilizer_spray_harvest_inventory_machinery_irrigation.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): High-resolution photos attached to a crop problem ticket.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores cloud image URL, capture timestamp, and visual symptom tags.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Enables computer vision retraining and gives human agronomists clinical visual evidence.

Harvest

`Table: harvests` 14 Fields · labour_fertilizer_spray_harvest_inventory_machinery_irrigation.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Records final yield and crop production output.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores harvest date, total quintals/bags, moisture percentage, plot area harvested, and storage location.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Measures farm productivity, yield per acre, and prepares stock for mandi sale.

HarvestGradeBreakdown

`Table: harvest_grade_breakdowns` 9 Fields · labour_fertilizer_spray_harvest_inventory_machinery_irrigation.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Quality grading breakdown of harvested produce (Grade A, Grade B, Rejection).

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores grain size, color, moisture, foreign matter percentage, and grade quintals.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Determines premium mandi pricing and fair valuation for grain buyers.

InventoryItem

`Table: inventory_items` 13 Fields · labour_fertilizer_spray_harvest_inventory_machinery_irrigation.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): On-farm inventory management for farmer's stored inputs and harvest.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Tracks item type (Seed, Fertilizer, Harvested Grain), quantity, unit, storage shed, and expiry date.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Prevents input wastage, spoilage, and helps plan procurement before market price rises.

InventoryTransaction

`Table: inventory_transactions` 14 Fields · labour_fertilizer_spray_harvest_inventory_machinery_irrigation.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Double-entry inventory ledger for all stock inward/outward movements.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Records transaction type (PURCHASE_IN, USAGE_OUT, SALE_OUT, WASTE), quantity, related plot/order.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Prevents inventory theft, provides audit trails, and feeds accurate cost-of-production numbers.

Machinery

`Table: machinery` 10 Fields · labour_fertilizer_spray_harvest_inventory_machinery_irrigation.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Registry of farm tractors, harvesters, rotavators, seed drills, and sprayers.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores model, horsepower, purchase date, fuel log, maintenance history, and hourly custom hiring rates.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Calculates machine depreciation, service schedules, and facilitates peer-to-peer machinery rental.

IrrigationRecord

`Table: irrigation_records` 17 Fields · labour_fertilizer_spray_harvest_inventory_machinery_irrigation.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Logs water application events across plots.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Records irrigation date, source (Tubewell/Canal/Drip), duration in hours, electricity/diesel units consumed.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Correlates with satellite moisture data to optimize water productivity and reduce pumping electricity bills.

UnifiedParty

`Table: unified_parties` 20 Fields · section_944.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Enterprise unified ledger merging trader, vendor, and buyer into a single legal counterparty.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores legal PAN, GSTIN, credit limit, and multi-currency/multi-account balances.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Supports large-scale FPO (Farmer Producer Organization) and corporate contract farming accounting.

ArhtiyaTransaction

`Table: arhtiya_transactions` 27 Fields · section_944.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Dedicated Punjab/Haryana APMC Mandi Arhtiya settlement tracker.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Records J-Form data, dami/commission rates, market fee, RD fee, RDF deductions, and net proceeds.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Ensures 100% compliance with APMC Act mandi statutory deductions.

Farm

`Table: farms` 23 Fields · users_farms_plots_crops.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Represents an agricultural holding or land estate owned/leased by a farmer.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Linked to User. Holds farm name, total acres, village, tehsil, district, state, soil type, and irrigation source.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Groups individual plots and establishes the geographic anchor for weather and satellite telemetry.

Plot

`Table: plots` 20 Fields · users_farms_plots_crops.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): A specific agricultural field or parcel within a farm.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores plot name, size in acres/kanals, polygon boundary geo-coordinates (GeoJSON), and current crop status.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Enables precise satellite NDVI scanning and personalized irrigation/fertilizer dosage calculation.

CropCycle

`Table: crop_cycles` 37 Fields · users_farms_plots_crops.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Records a single seasonal cultivation cycle (e.g. Rabi 2026 Wheat or Kharif 2026 Paddy).

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores crop variety, sowing date, expected harvest date, seed batch, acreage, current stage, and status.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Powers the automated crop calendar, pest alert timelines, and harvest yield records.

2. E-Commerce Marketplace & Multi-Vendor Inventory

This architectural pillar manages high-concurrency state transitions, relational data consistency, and operational workflows for **14** core database tables within the FarmsKing ecosystem.

Customer

`Table: customers` 12 Fields · expenses_sales_customers_payments.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Historical customer ledger tracking store buying behaviour, total spent, and loyalty points.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Linked to User with aggregate order counts, credit balance, and marketing preferences.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Powers personalized crop medicine recommendations and VIP discount tiers.

Product

`Table: products` 55 Fields · gardener_customer_business_partner_dashboards.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Main catalog of agricultural inputs (certified seeds, bio-fertilizers, organic pesticides, machinery).

2. How it works ("ਕਿਵੇਂ ਹੈ"): Contains title, description, SKU, HSN code, GST rate (0%, 5%, 12%, 18%), technical active ingredient, stock quantity, and sellerStoreId.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Provides transparent price discovery, statutory tax tags, and real-time inventory control across all multi-vendor sellers.

ProductReview

`Table: product_reviews` 10 Fields · gardener_customer_business_partner_dashboards.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Captures verified buyer feedback and ratings (1 to 5 stars) for agricultural inputs.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Linked to Product and User. Enforces verified purchase checks before accepting customer ratings and photo attachments.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Builds farmer trust, flags counterfeit/ineffective chemical formulations, and ranks high-quality sellers.

Wishlist

`Table: wishlists` 6 Fields · gardener_customer_business_partner_dashboards.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Enables farmers to bookmark seasonal fertilizers or tools before sowing seasons.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Simple junction table linking User and Product with unique constraints.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Allows targeted SMS/WhatsApp price-drop and seasonal stocking alerts.

CustomerOrder

`Table: customer_orders` 35 Fields · gardener_customer_business_partner_dashboards.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Master header record for all customer store purchases.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Tracks orderNumber (FK-ORD-XXXX), totalAmount, discountAmount, paymentMode (COD/ONLINE), paymentStatus (UNPAID/PAID), order status, and delivery address.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Initiates payment gateways, triggers invoice generation, routes items to sellers, and updates company escrow.

CustomerOrderItem

`Table: customer_order_items` 18 Fields · gardener_customer_business_partner_dashboards.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Itemized line items within a customer order.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores productId, sellerStoreId, unitPrice, quantity, GST rate, and status. Timestamped with createdAt and updatedAt.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Enables multi-vendor basket splitting so different sellers fulfill their respective products independently.

SellerStore

`Table: seller_stores` 57 Fields · section_192.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Verified commercial storefront for agrochemical manufacturers and distributors.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Holds storeName, GSTIN, PAN, bank account number, IFSC, category commission %, and KYC approval status.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Empowers direct seller logistics, payout calculations, and statutory GSTR-8 tax reporting.

SubOrder

`Table: sub_orders` 20 Fields · section_192.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Sub-division of a master order assigned to a specific seller store.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Grouped by sellerStoreId, tracking individual packing, labeling, and pickup status.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Allows independent dispatch, AWB generation, and settlement per seller.

ReturnRequest

`Table: return_requests` 14 Fields · section_192.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Manages damaged product returns, wrong seed deliveries, or transit leaks.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Captures orderItemId, reason, photo proofs, return courier tracking, and approval workflow.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Protects farmer purchase rights and enforces seller replacement SLAs.

SellerPayout

`Table: seller_payouts` 15 Fields · section_192.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Records commercial settlements disbursed to seller bank accounts.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Calculates gross sales minus platform fee, minus 18% GST on fee, minus 1% TCS u/s 52. Stores bank UTR.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Ensures full transparency, clean passbooks, and zero financial disputes with vendor partners.

ShippingLog

`Table: shipping_logs` 16 Fields · section_192.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Audit log of all courier interactions, AWB dispatches, and transit checkpoints.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores courier name (Shiprocket/BlueDart/Delhivery), AWB tracking ID, status changes, and webhook payloads.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Powers live order tracking screen and automated SMS delivery notifications.

Gstr8Report

`Table: gstr8_reports` 14 Fields · section_192.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Statutory monthly return for E-Commerce operators collecting 1% TCS under Section 52.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Aggregates net supplies, returns, and TCS amounts grouped by seller GSTIN for any calendar month.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Exportable format for filing on the Government GST Portal (gst.gov.in).

InvoiceSequence

`Table: invoice_sequences` 6 Fields · section_192.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Ensures zero gap and zero duplicates in financial year invoice numbering.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Atomic counter table per financial year with createdAt and updatedAt timestamps.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Strict adherence to Indian GST Rules for continuous sequential invoicing.

OrderInvoice

`Table: order_invoices` 25 Fields · section_192.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Legal statutory GST tax invoice generated for completed sales.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Contains unique sequential invoice number (e.g. FK-2627-0001), seller GSTIN, buyer billing details, HSN breakdown, CGST, SGST, IGST amounts, and printable PDF hash.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Mandatory compliance with Indian GST laws; downloadable by farmers for accounting.

9. Subscriptions, VIP Passes & Promotional Coupons

This architectural pillar manages high-concurrency state transitions, relational data consistency, and operational workflows for **16** core database tables within the FarmsKing ecosystem.

AdvisorPlan

`Table: advisor_plans` 12 Fields · farmer_advisor_plan_subscription_schedule_uploads_.prisma

1. Why it exists ("किउं है"): Service tier plan offered by crop doctors to farmers.

2. How it works ("किवें है"): Stores plan name, consultation count, fee, and validity.

3. Business & Data Impact ("इस नाल की होवेगा"): Standardizes doctor consultation offerings across the platform.

AdvisorSubscription

`Table: advisor_subscriptions` 19 Fields · farmer_advisor_plan_subscription_schedule_uploads_.prisma

1. Why it exists ("किउं है"): Active consultation subscription linking farmer to doctor under an AdvisorPlan.

2. How it works ("किवें है"): Tracks startDate, endDate, consultationsUsed, and payment request ID.

3. Business & Data Impact ("इस नाल की होवेगा"): Monitors remaining consultation quotas and enforces plan expiry.

PlanPaymentRequest

`Table: plan_payment_requests` 16 Fields · farmer_advisor_plan_subscription_schedule_uploads_.prisma

1. Why it exists ("किउं है"): Payment claim for generic subscription plans.

2. How it works ("किवें है"): Tracks amount, planId, status, and bank transaction reference.

3. Business & Data Impact ("इस नाल की होवेगा"): Manages subscription payment verification workflows.

FarmerPlanPaymentRequest

`Table: farmer_plan_payment_requests` 20 Fields · farmer_advisor_plan_subscription_schedule_uploads_.prisma

1. Why it exists ("किउं है"): Payment claim specifically for Farmer VIP passes and Option 3 upgrades.

2. How it works ("किवें है"): Stores farmerId, targetPlan, baseAmount, proratedCredit, finalAmount, utr, screenshotUrl, status, and confirmation metadata.

3. Business & Data Impact ("इस नाल की होवेगा"): Provides the auditable transaction bridge for manual UPI QR payments and automated gateway settlements.

FarmerPlan

`Table: farmer_plans` 11 Fields · section_378.prisma

1. Why it exists ("किउं है"): Currently active VIP software membership plan for a farmer.

2. How it works ("किवें है"): Tracks farmerId, plan (FREE/PRO/SMART/SUPER/ROYAL), startDate, endDate, and expiredAt.

3. Business & Data Impact ("इस नाल की होवेगा"): Determines feature gates across the mobile app (max crops, satellite radar access, doctor entitlement).

FarmerPlanHistory

`Table: farmer_plan_history` 16 Fields · section_378.prisma

1. Why it exists ("किउं है"): Historical log of all past plan activations, extensions, and prorated upgrades.

2. How it works ("किवें है"): Stores plan, status (ACTIVE/EXPIRED/SLEEP), daysGranted, remainingDays, notes, and timestamps.

3. Business & Data Impact ("इस नाल की होवेगा"): Maintains complete audit history of all plan transitions and Option 3 prorated credit adjustments.

AdvisorTierPlan

`Table: advisor_tier_plans` 8 Fields · section_378.prisma

1. Why it exists ("किउं है"): Internal classification of agronomist seniority tiers (Junior Doctor, Senior Consultant, Chief Agronomist).

2. How it works ("किवें है"): Stores tier name, required certifications, base payout rate, and supervisory privileges.

3. Business & Data Impact ("इस नाल की होवेगा"): Structures medical escalation hierarchies when complex crop epidemics arise.

FarmerPlanCoupon

`Table: farmer_plan_coupons` 33 Fields · section_378.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Redeemable VIP pass coupons distributed by business partners and doctors.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores unique King code (e.g. FP102948), plan, daysGranted, assignedFarmerId, isUsed, and commission splits.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Enables offline rural distribution where field partners sell physical scratch cards for cash.

FarmerPlanPricing

`Table: farmer_plan_pricing` 28 Fields · section_378.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Configurable master pricing table for farmer plans.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores plan tier, billingPeriodDays (e.g. 180, 365), base price, partner commission %, and doctor share %.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Allows Super Admins to adjust subscription pricing dynamically without modifying backend code.

GardenerPlan

`Table: gardener_plans` 13 Fields · section_703.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Active VIP membership plan for home gardeners.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Tracks gardenerId, plan tier, plant count limits, and validity.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Gates urban gardening features and advisory access.

GardenerPlanCoupon

`Table: gardener_plan_coupons` 15 Fields · section_703.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Promotional discount cards for urban gardening kits and plants.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores code, discount %, validity, and redemption status.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Powers urban nursery promotions and landscape marketing campaigns.

Coupon

`Table: coupons` 21 Fields · section_860.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): E-Commerce discount vouchers applied during store cart checkout.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores couponCode, discountType (PERCENTAGE/FIXED), discountValue, minOrderValue, maxDiscount, and validity.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Incentivizes larger basket sizes and boosts seasonal input sales.

CouponRedemption

`Table: coupon_redemptions` 12 Fields · section_860.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Tracks individual usages of checkout discount coupons.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores couponId, userId, orderId, and discountAmountApplied.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Enforces per-user usage limits and calculates total marketing promotion expenditure.

CouponSystemSetting

`Table: coupon_system_settings` 24 Fields · section_860.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Global configuration setting default commission shares and referral rates for coupons.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores defaultPartnerSharePercent, defaultDoctorSharePercent, minWithdrawalLimit, and tax rules.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Central control knob for platform unit economics and affiliate payouts.

PlanRenewalCoupon

`Table: plan_renewal_coupons` 13 Fields · section_860.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Coupons locked to specific farmers or advisors to extend expiring memberships.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores code, daysGranted, assignedFarmerId, assignedAdvisorId, and usage timestamp.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Prevents churn by enabling doctors to gift extension days to loyal farmers.

BasicPlanCoupon `Table: basic_plan_coupons` 15 Fields · section_860.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Entry-level promotional vouchers unlocking trial access.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores code, trialDays, maxRedemptions, and expiry date.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Drives viral user acquisition in new agricultural districts.

11. Telecommunications & Push Messaging

This architectural pillar manages high-concurrency state transitions, relational data consistency, and operational workflows for **4** core database tables within the FarmsKing ecosystem.

Upload

`Table: uploads` 8 Fields · farmer_advisor_plan_subscription_schedule_uploads_.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Master registry of all cloud files, images, PDFs, and invoices uploaded to platform.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores storageKey, originalFilename, mimeType, fileSize, uploaderId, and public URL.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Maintains strict file storage security and prevents orphaned unlinked media.

Notification

`Table: notifications` 10 Fields · market_rates_notifications.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): In-app notification inbox delivered to users.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores userId, title, message body, category (SYSTEM, ORDER, PAYMENT, AGRONOMY), read status, and action URL.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Keeps farmers informed of weather warnings, order dispatches, and doctor replies in real-time.

AdminChatMessage

`Table: admin_chat_messages` 11 Fields · section_703.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Direct support chat between platform users and central administrative desk.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores userId, adminId, message text, priority, and resolution status.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Delivers immediate customer support and solves order/payment grievances.

Message

`Table: messages` 9 Fields · section_860.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Peer-to-peer real-time chat messages between farmers and crop doctors.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores senderId, receiverId, message text, attachment URL, read status, and message timestamps.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Enables instant agronomic troubleshooting and photo sharing directly inside the app.

6. Urban Horticulture & Garden Advisory

This architectural pillar manages high-concurrency state transitions, relational data consistency, and operational workflows for **7** core database tables within the FarmsKing ecosystem.

Garden

`Table: gardens` 13 Fields · gardener_customer_business_partner_dashboards.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Represents an urban home garden, terrace garden, or commercial landscape lawn.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores garden name, square feet, sunlight exposure (Full/Partial), soil type, and location.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Extends FarmsKing ecosystem to urban plant parents and terrace farming enthusiasts.

Plant

`Table: plants` 11 Fields · gardener_customer_business_partner_dashboards.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Individual potted plant, fruit tree, or ornamental shrub within a garden.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores plant species, botanical name, potting date, health status, watering interval, and custom notes.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Generates tailored watering, fertilizing, and pruning reminders for home gardeners.

GardenVisit

`Table: garden_visits` 8 Fields · gardener_customer_business_partner_dashboards.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Physical on-site inspection visit conducted by a certified Garden Expert.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores scheduled date, expertId, gardenId, checklist completed, soil test results, and visit report.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Delivers premium doorstep plant health inspections in urban metropolitan sectors.

GardenExpense

`Table: garden_expenses` 14 Fields · section_309.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Expenditure tracker for urban pots, potting mix, organic manure, and decorative plants.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Tracks date, amount, category, merchant name, and receipt attachment.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Helps garden owners budget their annual landscaping and plant care costs.

GardenExpertPlan

`Table: garden_expert_plans` 11 Fields · section_309.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Commercial service tier for urban plant maintenance (Monthly Care, Seasonal Pruning).

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores plan name, price, visits included, free repotting quota, and billing period.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Generates recurring subscription revenue from urban households.

GardenExpertAssignment

`Table: garden_expert_assignments` 15 Fields · section_309.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Links a urban gardener to their dedicated plant doctor.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Tracks expertId, gardenerId, planId, status, and service start/end dates.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Ensures accountability and seamless communication for urban landscaping clients.

GardenerPlanPaymentRequest

`Table: gardenerplanpaymentrequest` 8 Fields · section_309.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Payment claim for garden advisory plan subscriptions.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores amount, planId, payment status, UTR, and confirmation timestamps.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Manages subscription collection and routes payouts to garden experts.

4. Satellite Telemetry & Weather

This architectural pillar manages high-concurrency state transitions, relational data consistency, and operational workflows for **5** core database tables within the FarmsKing ecosystem.

MarketRate

`Table: market_rates` 13 Fields · market_rates_notifications.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Daily mandi commodity price discovery engine.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Pulls official Agmarknet data across APMC mandis for Wheat, Basmati, Cotton, Mustard, Potato, etc. Stores min, max, modal prices.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Prevents distress selling by informing farmers of real-time price trends before taking grain to the market.

PlotSatelliteScan

`Table: plot_satellite_scans` 9 Fields · section_381.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Historical log of multispectral Sentinel-2 satellite passes over a farmer's plot.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores scan date, cloud coverage %, mean NDVI score, NDWI soil moisture score, and raster tile URL.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Builds a multi-month vegetation health curve, proving crop vigor and validating crop insurance claims.

MandiPricePrediction

`Table: mandi_price_predictions` 7 Fields · section_381.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): AI price forecasting engine estimating commodity prices 7-30 days in advance.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Trained on historical seasonal arrivals, weather trends, and domestic consumption patterns.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Guides farmers whether to sell immediately or store produce in warehouse for higher future prices.

PriceLockContract

`Table: price_lock_contracts` 11 Fields · section_381.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Digital forward contract locking a guaranteed purchase price between farmer and buyer.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores commodity, grade, agreed price, delivery deadline, buyer earnest deposit, and contract hash.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Protects farmers against sudden market price crashes during peak harvesting weeks.

WeatherTipTemplate

`Table: weather_tip_templates` 10 Fields · section_860.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Localized agro-meteorological advisory rules triggered by forecast conditions.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Maps weather events (Rain >20mm, Frost <4°C, High Wind >35km/h) to practical agronomic advisories.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Warns farmers not to spray chemicals before heavy rainfall, saving thousands in washed-away chemicals.

12. Statutory Audit, ISO 27001 Security & System Governance

This architectural pillar manages high-concurrency state transitions, relational data consistency, and operational workflows for **7** core database tables within the FarmsKing ecosystem.

PartnerReferral

`Table: partner_referrals` 7 Fields · section_192.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Commercial referral linkage for institutional business partners and agri-input retail stores.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Maps retail store referrals to onboarded farmers, calculating monthly wholesale incentives.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Integrates traditional village fertilizer dealers into the FarmsKing digital distribution channel.

IsoModuleControl

`Table: iso_module_controls` 9 Fields · section_309.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Dynamic operational kill-switch and maintenance mode controls for platform modules.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Allows Super Admins to lock or enable specific features (e.g. Call Requests, Store Checkout, Satellite Scanning) globally.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Enables graceful maintenance without breaking the entire platform.

IsoAuditLog

`Table: iso_audit_logs` 7 Fields · section_309.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): ISO 9001 quality management log tracking platform SLA adherence and uptime metrics.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Records server health, incident response times, and quality audit sign-offs.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Provides institutional statutory certification for government tenders and enterprise contracts.

AuditLog

`Table: audit_logs` 11 Fields · section_860.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Immutable audit trail of every data mutation executed on the platform.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Captured automatically by Global AuditLogInterceptor. Stores actorId, actorKingId, ipAddress, endpoint, httpMethod, statusCode, and sanitized payload diff.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Guarantees complete non-repudiation and complies with ISO 27001 and IT Act 2000 forensic standards.

AppSetting

`Table: app_settings` 42 Fields · section_860.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Central configuration store for corporate master data, API keys, statutory numbers, and gateway credentials.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Single master row holding company legal name, CIN, PAN, TAN, GSTIN, registered office address, support phone, UPI ID, free trial toggles, and satellite API keys.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Enables zero-code modification of statutory details and payment routes from Admin Console.

UserReferral

`Table: user_referrals` 9 Fields · section_890.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Tracks viral grassroots affiliate signups and referral uplines.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores referrerId, refereeId, referralCode, and status (PENDING/CONFIRMED).

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Rewards progressive farmers for introducing neighboring villages to the platform.

ReferralRoyalty `Table: referral_royalties` 9 Fields · section_890.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Calculates multi-tier residual commission paid to top rural influencers.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores recipientId, sourceOrderId/PlanId, tierLevel (1, 2, 3), royalty amount, and disbursement status.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Creates an ongoing secondary income stream for rural village promoters.

7. Kisan Academy & Agronomy Training

This architectural pillar manages high-concurrency state transitions, relational data consistency, and operational workflows for **2** core database tables within the FarmsKing ecosystem.

TrainerAssignment

`Table: trainer_assignments` 17 Fields · section_309.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Assigns a certified agricultural master trainer to a cohort of farmers or village territory.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Tracks trainerId, farmerId, module curriculum, assignment date, and completion milestones.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Drives structured capacity building in organic farming, beekeeping, and high-density orchards.

FarmerTrainingLog

`Table: farmer_training_logs` 24 Fields · section_309.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Logs progress of farmers completing video courses and practical training modules.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores courseId, lessonId, quiz score, hours attended, and certificate issuance status.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Issues verified farmer credentials that unlock government subsidies and bank loans.

1. Core Identity & Multi-Role RBAC

This architectural pillar manages high-concurrency state transitions, relational data consistency, and operational workflows for **3** core database tables within the FarmsKing ecosystem.

ExpertApplication

`Table: expert_applications` 19 Fields · section_309.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Governs formal accreditation for agricultural doctors, soil scientists, and trainers applying to join platform.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores applicant resume, degree certificates, university accreditation, years of experience, and admin review status (PENDING/APPROVED/REJECTED).

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Ensures only verified agronomists and ICAR-certified specialists advise farmers.

CustomerAddress

`Table: customer_addresses` 13 Fields · section_860.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Stores multiple delivery destinations (Home, Village Farm, Warehouse) for e-commerce shipments.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Linked to User via userId. Captures street, village, district, state, pin code, GPS coordinates, and default address flag.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Enables one-click checkout and automated Shiprocket courier pincode serviceability checks.

User

`Table: users` 208 Fields · users_farms_plots_crops.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Single unified record for every human/business actor on FarmsKing (Farmer, Doctor, Seller, Customer, Trainer, Admin). Eliminates fragmented accounts.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Primary table with unique King ID, mobile number, Argon2 password hash, role enum, and roles array. Holds dynamic supervisor configs, KYC flags, and profile fields.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Unlocks role-based dashboards dynamically. A farmer can buy inputs, sell produce, and consult doctors under a single identity.

10. King Connect B2B Mandi Shared Ledger

This architectural pillar manages high-concurrency state transitions, relational data consistency, and operational workflows for **4** core database tables within the FarmsKing ecosystem.

KingConnectLink

`Table: king_connect_links` 15 Fields · section_636.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Digital relationship link established between a farmer and a grain trader/buyer.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores farmerId, buyerId, linkStatus (ACTIVE/BLOCKED), trade terms, and credit limits.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Forms the trust baseline for direct farmer-to-buyer grain trade without middleman extortion.

P2pLedgerSyncRequest

`Table: p2p_ledger_sync_requests` 22 Fields · section_636.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Synchronization request ensuring farmer's ledger and buyer's ledger match down to the rupee.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Compares transaction hashes, records discrepancies, and demands reciprocal electronic sign-off.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Produces a court-admissible shared accounting statement during trade disputes.

DemandRequest

`Table: demand_requests` 21 Fields · section_636.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Mandi procurement demand posted by grain mills, exporters, or buyers.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores commodity, required quintals, target purchase price, delivery mandi location, and expiry date.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Gives farmers direct visibility into institutional wholesale demand before harvesting.

KingPaymentRequest

`Table: king_payment_requests` 19 Fields · section_636.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Commercial payment request generated after mutual grain delivery confirmation.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores linkId, amount, dual signatures, escrow status, and release approval flags.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Eliminates payment default risks in inter-state grain transactions.

8. Digital Wallets, Payouts & Escrow Accounting

This architectural pillar manages high-concurrency state transitions, relational data consistency, and operational workflows for **2** core database tables within the FarmsKing ecosystem.

WalletTransaction

`Table: wallet_transactions` 14 Fields · section_860.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Double-entry append-only ledger for all wallet balances.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Tracks walletId, amount, type (CREDIT/DEBIT), category (ORDER_PAYMENT, REFERRAL_BONUS, SELLER_PAYOUT), balanceAfter, and reference ID.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Guarantees zero balance discrepancy and mathematically proves every rupee entering or leaving platform.

WithdrawalRequest

`Table: withdrawal_requests` 15 Fields · section_860.prisma

1. Why it exists ("ਕਿਉਂ ਹੈ"): Formal bank transfer request submitted by sellers, doctors, or farmers withdrawing wallet funds.

2. How it works ("ਕਿਵੇਂ ਹੈ"): Stores amount, userId, bankAccountNumber, IFSC, status (PENDING/CONFIRMED/REJECTED), statutory TDS deducted, and bank UTR.

3. Business & Data Impact ("ਇਸ ਨਾਲ ਕੀ ਹੋਵੇਗਾ"): Ensures full compliance with banking KYC and handles batch payout disbursements securely.

Developer Operational Guidelines & Statutory Compliance

Golden System Commandments

Commandment 1: 100% Pure English User Interface & Documentation

All user-facing UI screens, dashboards, buttons, telemetry badges, and documentation must strictly adhere to 100% professional English with easy wording. Vernacular hardcoded subtitles are strictly prohibited (the only exception is the Dedicated Language Switcher Box).

Commandment 2: Primary PC/Laptop Screen Standard

All features must be engineered, verified, and visually benchmarked on PC/Laptop web displays first. Mobile view adaptation executes cleanly in the background.

Commandment 3: Zero Automatic Live Deployments

Never run automated deployment routines to the live production server (Hostinger) without explicit user authorization ("live deploy kro"). All testing executes locally on Ports 3000, 8081, and 8082.

Commandment 4: Option 3 Plan Upgrades & Strict Downgrade Protection

Lower-tier to higher-tier upgrades apply Prorated Unused Credit Adjustment. Downgrading from an active higher-tier plan to a lower-tier plan is strictly prohibited and blocked by both backend APIs and frontend UI.

System Health & Verification Summary

SUBSYSTEM	CONFIGURATION / PATH	OPERATIONAL STATUS
Relational Database	103 Tables (Neon PostgreSQL Serverless)	100% Active · All models equipped with createdAt & updatedAt
Backend REST API	`backend/src/main.ts` (Port 3000)	Verified Clean · 0 TypeScript compilation errors
Farmer Web App	`frontend/` (Expo React Native Web)	Bundled Cleanly (8.2MB Web Bundle)
Admin Console	`admin/` (Expo React Native Web)	Bundled Cleanly (7.4MB Web Bundle)
Audit Interceptor	`AuditLogInterceptor` in `backend/src/app.module.ts`	Active across all POST, PUT, PATCH, DELETE mutations

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